

error_10_mape_vs_wmape.Rmd

```
x <- seq(0, 100000, length.out = 500000)

y <- x

upper_mape <- 1.1 * x
lower_mape <- 0.9 * x

mean_x <- mean(x)
constant_error <- 0.10 * mean_x

upper_wmape <- x + constant_error
lower_wmape <- x - constant_error

plot(x, y, type = "l", lwd = 3,
     xlab = "Actual (A)",
     ylab = "Prediction (P)",
     main = "10% Error: MAPE vs wMAPE",
     ylim = c(-500, 110000))

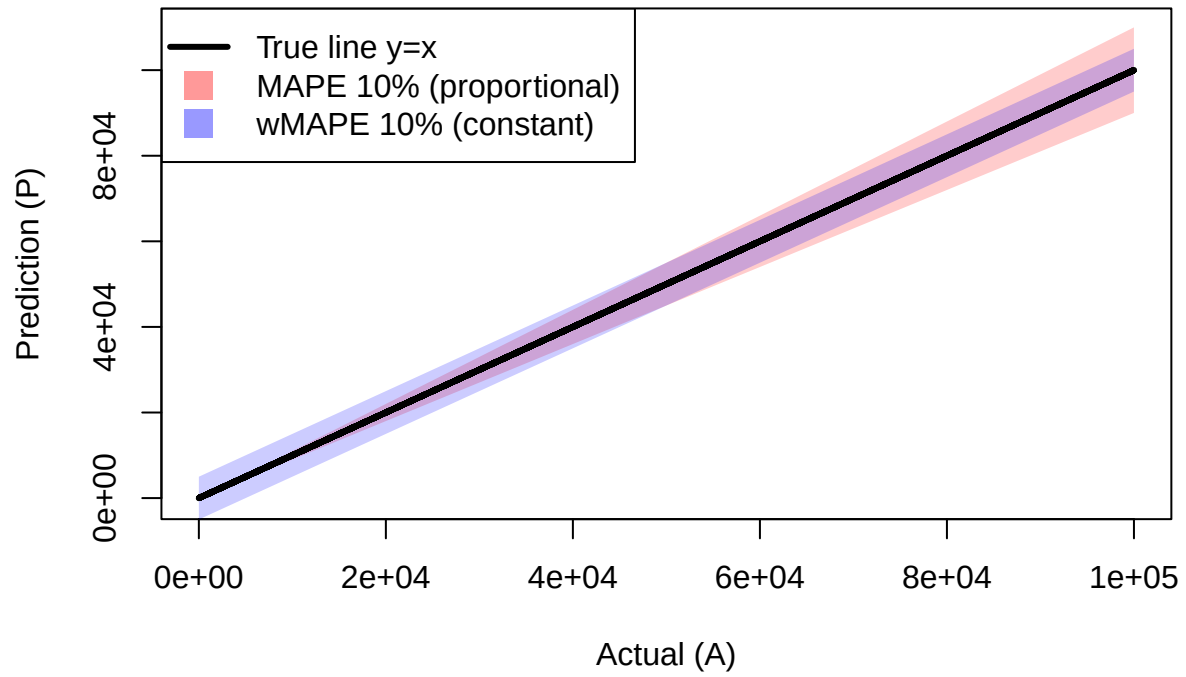
polygon(c(x, rev(x)),
       c(upper_mape, rev(lower_mape)),
       col = rgb(1, 0, 0, 0.2), border = NA)

polygon(c(x, rev(x)),
       c(upper_wmape, rev(lower_wmape)),
       col = rgb(0, 0, 1, 0.2), border = NA)

lines(x, y, lwd = 3)

legend("topleft",
     legend = c("True line y=x",
                 "MAPE 10% (proportional)",
                 "wMAPE 10% (constant)"),
     lwd = c(3, NA, NA),
     pch = c(NA, 15, 15),
     pt.cex = 2,
     col = c("black",
             rgb(1,0,0,0.4),
             rgb(0,0,1,0.4)))
```

10% Error: MAPE vs wMAPE



```
col = rgb(1,0,0,0.2)
col = rgb(0,0,1,0.2)
lwd = 3
```